

Animesh Sharma

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EDUCATION

Parul University

B.Tech in Computer Science & Engineering | CGPA: 7.83

Vadodara, Gujarat

2023 – 2027

EXPERIENCE

Open Source Contributor

May 2026

GirlScript Summer of Code (GSSoC) — CommitPulse (GitHub analytics platform)

- Fixed an unbounded **TTLCache** OOM leak (`lib/cache.ts`) caused by lazy-only eviction; added active 60s GC sweep, FIFO **maxSize** eviction, `timer.unref()` process protection, and a **destroy()** lifecycle with 8 unit tests covering fake-timer sweep validation.
- Replaced **Promise.all** with **Promise.allSettled** in the GitHub API fetch layer, isolating endpoint failures so a rate-limit on one fetch no longer crashes the full dashboard; added **AbortSignal** for hanging requests and typed fallbacks to guard `calculateStreak`.

Freelance Software Engineer

Feb 2026

Hangout Spot — Cafe Billing & POS System | Flutter, Firebase, Dart

- Replaced POS (Rs. 12,000 first year + Rs. 7,000–10,000/year renewal) with a zero-renewal Flutter app; **eliminated 100% of recurring SaaS costs** as Firebase handles cafe-scale peak traffic within free-tier limits.
- Built a **dual-database architecture** (SQLite/Drift offline + Firebase Realtime DB) enabling **sub-50ms** live analytics and atomic multi-device cart sync with conflict resolution.
- Implemented atomic invoice generation preventing duplicates across concurrent devices, Bluetooth thermal printing via ESC/POS, and role-based access control for staff.

PROJECTS

Parallax — Collaborative Development Platform | React, TypeScript, Spring Boot, WebRTC, Docker

[GitHub]

- Built a full-stack collaborative IDE supporting **50+ concurrent users** with OT-inspired sync, live cursors, and multi-file editing via Monaco Editor.
- Integrated **WebRTC** with a custom STOMP/WebSocket signaling engine for in-IDE peer-to-peer voice and video collaboration.
- Engineered **Docker-orchestrated isolated code execution** sandboxing untrusted runtimes, achieving **sub-150ms** cold-start latency using warm container pre-allocation.

NextLevel — Productivity Platform | Spring Boot, Spring AI, MongoDB Atlas Vector Search

[Live] [GitHub]

- Architected a semantic knowledge vault using **Spring AI**, **Groq Llama-3.3**, and MongoDB Atlas Vector Search (768-dimensional embeddings), enabling concept-based retrieval beyond keyword matching through cosine similarity ranking.
- Engineered a fault-tolerant AI ingestion pipeline using **JobRunr** and **Resilience4j** with exponential backoff, asynchronously scraping, summarizing, and vectorizing saved content while recovering automatically from transient LLM API failures and rate limits.
- Built a real-time **RAG conversational assistant** with Server-Sent Events (SSE) streaming, alongside a custom **SM-2 spaced repetition engine** and gamification system to optimize personalized learning and productivity workflows.

JSON Parser Library | Java, JUnit 5

[GitHub]

- Built a **zero-dependency** Java library implementing a custom lexer and recursive descent parser to parse, query, validate, and diff JSON schemas.
- Engineered an event-driven streaming parser using a custom state machine to process **10MB+** nested payloads in **O(1)** auxiliary space, reducing peak heap allocation by **>95%**.
- Implemented **JSONPath-like query syntax** with O(1) path indexing of deep nodes and column-level syntax error diagnostics.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, Dart, C, SQL

Backend: Spring Boot, Spring MVC, Spring Security, Hibernate, JPA, Apache Kafka, Redis, REST APIs, JWT, WebSockets

Frontend: React, Vite, Tailwind CSS, WebRTC, Monaco Editor

AI/ML: Spring AI, PgVector, LLM Integration (OpenAI API), Machine Learning

Cloud / DevOps: Docker, AWS, Firebase, Git

Databases: PostgreSQL, MongoDB, MySQL, SQLite

CERTIFICATIONS & ACHIEVEMENTS

- ServiceNow Certified System Administrator (CSA).
- HackerRank Certified: Java & Python.
- Selected for **State-Level Regional** at OpenAI Academy × NxtWave × IndiaAI Hackathon.
- Solved **700+ DSA problems**